Manoj N H

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Education

Indian Institute of Technology Madras (IITM) 2023 - 2027

Bachelor of Technology in Electrical Engineering *(GPA: 8.71)*
Minor in Neuroscience *(Minor GPA#: 10.00)* & Physics *(Minor GPA#: 9.00)*

Selected Coursework:

EE6347# (Devices & Technologies for AI & Neuromorphic Computing)
CS6251# (Computational Models of Cognition)
EE6412 (Optimal Control)
AM5700 (Mechanistic and Data-Driven Modelling of Complex Systems)
BT6270 (Computational Neuroscience)

Sri Chaitanya School, JP Nagar, Bengaluru 2023

Highschool; Grade 10 (AISSE: 99.2%), Grade 12 (AISSCE: 95.0%)

Scholastic Achievements

- Secured an **All-India Rank (AIR) 559** in the **JEE Advanced Exam 2023** out of 190k candidates.
- Scored **100 percentile** in the Physics section and ranked in **top 0.165%** overall in **JEE-Mains 2023.**
- Secured **AIR 53** in the **IISER Aptitude Test,** used for admissions to top science programs at IISc, IISERs.
- Recipient of the **National Talent Search Examination (NTSE) Fellowship** by NCERT, Govt. of India.
- Selected for a 5-day **Nurturance Program for NTSE fellows** conducted by IISc Bangalore in Dec 2022.
- Accepted into an International two-week program on 'Unifying Theories in High-Dimensional Biophysics' held at ICTS-Bengaluru, in July 2025, aimed primarily at **graduate students.**

Research Experience

Interface Lab, IITM
Undergraduate Researcher
March '25 - Present

Emergence of Innovation in Collectives, with Prof. Ganga Prasath, IITM

- Formulated insect navigation as a continuous-space **optimal control** problem with behavioral switching, coupling Hamilton–Jacobi–Bellman (HJB) value function dynamics with **Fokker–Planck equations** for collective agent densities.
- Used a Cole–Hopf transformation to linearize the HJB dynamics, showing that the **collective value function manifests as a pheromone;** this mathematical structure underlies stochastic control, score-based diffusion/generative models.
- Connected **pheromone evaporation to temporal discounting** and validated the continuum formulation with agentic simulations, demonstrating trail formation.

Fiete Lab, MIT
Undergraduate Researcher (Remote)
Oct '25 - Present

Synaptic Patterning in Heterosynaptic Plasticity, with Prof. Ila Fiete, MIT

- Modeling how competition in heterosynaptic plasticity can establish **spatial patterning of synaptic weights** along the length of a dendritic branch, offering an explanation of observed clustering of spines with similar strengths.
- Aiming to demonstrate how these heterosynaptic interactions stabilize synaptic strengths against noise and create **longer-lived collective states**, even though individual synapses fluctuate and remodel on much faster timescales.
- Exploring if such interactions can provide an unsupervised mechanism against **catastrophic forgetting** by preserving learned representations while still allowing the plasticity required for new learning (**stability–flexibility tradeoff**).

#- Ongoing

Computational Neuroscience Lab, IITM

Project Intern

Dec '24 - Jan '25

Continuous Learning (CL) in Machines and Humans

- Conducted a literature review on **catastrophic forgetting** in sequential Atari tasks, focusing on mechanisms to mitigate interference in continual learning.
- Reviewed research on the use of Atari games for **motor & cognitive rehabilitation**, with emphasis on engagement, task difficulty, & feedback.
- Concluded on how **continual-learning strategies** may support adaptive training regimes for stroke rehabilitation, where patients must relearn tasks without overwriting prior motor skills.

Selected Projects

Goedel Machines

Robotics Intern

Jan'26 - Jul'26

- Investigated how **reward structure shapes emergent behavior in RL**, analyzing reward hacking across locomotion tasks (walking, jumping, and hindstanding).
- Designed a **behavioral manager** for state-dependent switching between learned policies, enabling adaptive locomotion across changing tasks and environments.
- Developed the **Isaac Lab + RSL-RL** training and **sim-to-real** pipeline for Unitree Go2, using terrain randomization and perturbations for robustness; extended it to a **custom-built quadruped** to study cross-platform policy transfer.

Protein Folding with Quantum Algorithms

Course Project, ID5841: Quantum Computing Lab

- Reviewed literature on approaches to Protein Structure Prediction, most relying on lattice approximations of 3D structures, and avenues for quantum speedup.
- **Reformulated** a recent method from literature (Fakhoury et. al., 2024) on Contact-Map based Protein Folding as a novel Quadratic Unconstrained Binary Optimization (**QUBO**) problem, without making lattice approximations.
- Solved the resulting QUBO using Quantum Annealing with **$O(n^2)$ scaling**, comparing solution quality and against simulated annealing on test sequences.

Selected Positions of Responsibility (IITM)

Biotech Club, Centre for Innovation (CFI)

Project Lead for Neurospike

May'24 - Present

Project Member, Neurospike

Jun'24 - Mar'25

- **Lead** a 5-member interdisciplinary team exploring **neuromorphic learning systems** on hardware; defined the project scope, led team member recruitment.
- In the first phase, reviewed **Kuramoto-oscillators** and implemented a pre-trained **SNN on an FPGA**, benchmarking latency and energy efficiency relative to ANNs.
- Explored approaches to solve the training problem for deep SNNs, investigating **self-learning** approaches such as **Predictive Coding** to overcome limitations of gradient-based methods and extend beyond what STDP alone can support.
- Designed and tested hierarchical predictive-coding modules with the long-term goal of enabling online, low-power learning in deeper spiking architectures.

Extracurriculars

- **Winner ('23)** and **Third ('24)** of 'a_i to AI', a contest focused on rigorously testing the **math behind AI.**
- **3rd Place,** IEEE-ISIT Q. Hackathon '25 (U. Michigan): Developed an algorithm using Quantum Annealing to optimize beamforming in Intelligent Reflecting Surface (IRS) with applications in 6G communication.
- **Organizer & Mentor,** Qiskit Fall Fest '25 (Sponsored by IBM): Handled logistics and planning for guest lectures and mentored participants in a hackathon. Held a lecture on Interaction-Free measurements.
- **Co-founded** a Social Impact campaign, 'Empower', for people with physical disabilities through Saarang (IITM's annual cultural fest); received **UNESCO Patronage** and reached an audience of **100k**.
- **National level Semi-Finalist** in the NTPC Electron Quiz '25; **15+ podium finishes** in collegiate and open quiz contests nationwide (including the Inter-IIT Quiz Cup). Organized quizzes on behalf of IITM.
- **Ranked 2nd** in a 24-hour hackathon (by GeeksforGeeks x TechSoc-IITM) on time series prediction of cryptocurrency coin prices based on sentiment data from tweets, using **multi-modal AI models.**
- **Winner** of the 'Anatomy of a Murder', conducted by the iGEM team, IITM, on analyzing chemical synthesis pathways in genetically modified organisms & optimizing target drug synthesis strategies.